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NMR relaxometric probing of ionic liquid dynamics and diffusion under mesoscopic confinement within bacterial cellulose ionogels

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Bacterial cellulose ionogels (BCIGs) represent a new class of material comprising a significant content of entrapped ionic liquid (IL) within a porous network formed from crystalline cellulose microfibrils. BCIGs suggest unique opportunities in separations, optically active materials, solid electrolytes, and drug delivery due to the fact that they can contain as much as 99% of an IL phase by weight, coupled with an inherent flexibility, high optical transparency, and the ability to control ionogel cross-sectional shape and size. To allow for the tailoring of BCIGs for a multitude of applications, it is necessary to better understand the underlying principles of the mesoscopic confinement within these ionogels. Toward this, we present a study of the structural, relaxation, and diffusional properties of the ILs, 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([emim][Tf₂N]) and 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide ([bmpy][Tf₂N]), using ¹H and ¹⁹F NMR T₁ relaxation times, rotational correlation times, and diffusion ordered spectroscopy (DOSY) diffusion coefficients, accompanied by molecular dynamics (MD) simulations. We observed that the cation methyl groups in both ILs were primary points of interaction with the cellulose chains and, while the pore size in cellulose is rather large, [emim]⁺ diffusion was slowed by ~2-fold, whereas [Tf₂N][−] diffusion was unencumbered by incorporation in the ionogel. While MD simulations of [bmpy][Tf₂N] confinement at the interface showed a diffusion coefficient decrease roughly 3-fold compared to the bulk liquid, DOSY measurements did not reveal any significant changes in diffusion. This suggests that the [bmpy][Tf₂N] alkyl chains dominate diffusion through formation of apolar domains. This is in contrast to [emim][Tf₂N] where delocalized charge appears to preclude apolar domain formation, allowing interfacial effects to be manifested at a longer range in [emim][Tf₂N]. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5016337>

I. INTRODUCTION

Ionic liquids (ILs) have utility for a wide array of applications, including electrochemical devices,¹ reaction media,^{2,3} pharmaceuticals,^{4,5} gas separation or capture platforms,^{4,6,7} and sensory materials,^{8,9} which has driven endeavors to understand their physical and chemical properties (e.g., viscosity, density, thermal stability, solubility, structure, ionic conductivity, and diffusion coefficients).^{10–15} Understanding these properties has allowed the IL community to further their use in applications by understanding the molecular level details that govern their behavior, as functions of variable pressure and temperature, for example.

Given the liquid nature of ILs, a number of practical impediments can complicate their use in actual devices.^{16–18} These issues have been addressed by IL immobilization by various techniques that include covalent attachment of IL cations or anions to solid supports,^{18–21} formation of thin

liquid films adsorbed to resins,²² confinement within porous structures,²³ or polymerization of reactive monomers to generate polymer analogs of ionic liquids (i.e., poly-ILs).²⁴ In addition, the use of ionogels²⁵ has also become a popular choice for generating self-supported, monolithic layers. Ionogels often maintain or potentially enhance attractive IL properties (e.g., reduction in melting temperature) while conversely enhancing or stabilizing the supporting matrix in some cases.^{23,26,27} Silica ionogels prepared from tetramethoxysilane (TMOS) and 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([bmim][Tf₂N]) using a 1:1 molar ratio displayed transition temperatures that were reduced by 20 K in differential scanning calorimetry (DSC) experiments, yet all ionogels presented ionic conductivities of the same order of magnitude and thermal stabilities similar to that of neat [bmim][Tf₂N].²⁸ Watanabe and co-workers showed that ionogels prepared from poly(methylmethacrylate) (PMMA) and [emim][Tf₂N] provided higher degradation temperatures for PMMA (by an additional 50 K) and improved conductivity as well, with increased [emim][Tf₂N] content ionogels becoming bulk-like in behavior at ~90 wt. % [emim][Tf₂N].²⁹

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Many other examples exist, especially in silica ionogels,^{23,30} which has spurred activity in ionogel research. While techniques used to study thermal properties are important for elucidating stability and thermal transitions, an in-depth microscopic understanding of these materials is better illuminated by techniques such as NMR,^{16,31–34} molecular dynamics (MD) simulations,^{34–36} and quasi-elastic neutron scattering (QENS).^{37–40} QENS has been used to study diffusion within a biopolymer/inorganic composite (methyl cellulose/silica) ionogel containing up to 91 wt. % IL.³⁸ The [bmim][Li][Tf₂N] mixture was found to exhibit diminished diffusion, with several distinct populations of IL due to multiple surfaces of interaction. PMMA ionogels up to 27 mol. % [emim][Tf₂N] showed two types of dynamical motion: one slower motion due to matrix coupling and a faster translational motion associated with the neat IL.³⁹ Other studies have been performed in porous carbon and silica, which also revealed changes to the diffusive motions compared to that of the neat liquid.^{23,27,30,37,40,41}

NMR spectroscopy has been widely used to study the dynamics and structural motifs of ionogels. A study of [bmim][Tf₂N] confined in silica ionogels showed spin-lattice (T_1) relaxation times that were slower than in bulk IL and did not follow the $T_1 = T_2$ trend within the high-temperature regime of relaxation measurements where spin-spin (T_2) relaxation times were one to two orders of magnitude shorter.³² Rotational correlation times (τ_c) calculated from the imidazolium side chain T_1 relaxation times showed segmented motion and were longest for the cation methyl group.³² In other neat liquids, such as 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide,⁴² [bmpy][Tf₂N], were also found to show similar segmental motions in the reorientation dynamics and were observed to have slow N—CH₃ rotations, which changed depending on the cation structure.^{16,43–45}

Complementing the relaxation measurements, the self-diffusion coefficients for ILs have been measured in solid polymer electrolytes, neat ILs, and ionogels to probe dynamical processes using pulsed field gradient (PFG) NMR techniques. Neat imidazolium and pyrrolidinium ILs studied by diffusion ordered spectroscopy (DOSY)-NMR were reported to have different cation and anion interactions based on the differences between the self-diffusion coefficients of the ¹H (seen in the cation) and ¹⁹F (seen in the anion) nuclei.⁴⁶ To illustrate the differences, D values for [emim][Tf₂N] and 1-propyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide, [pmpy][Tf₂N], at 298 K were 5.04×10^{-11} and 2.90×10^{-11} m² s⁻¹ in the cations, respectively, and the anions had D values of 2.94×10^{-11} and 2.20×10^{-11} m² s⁻¹, respectively.⁴⁶ When considering the cation and anion ratios of each liquid, the difference suggests that the IL ion diffusion depends greatly on the ionic interaction. Hagaman and colleagues³² showed that silica ionogels containing various amounts of [bmim][Tf₂N] were shown to exhibit an order of magnitude difference in diffusional properties within fully filled pores and pores with only enough IL to cover the surface. By understanding the underlying driving forces, Martinelli and co-workers⁴⁷ achieved conductivities and mobilities in silica ionogel micro-particles that were improved over the standard silica ionogel using

a more hydrophobic surface. PMMA-based 90 wt. % ionogels have also been shown to reduce the diffusional dynamics of the [emim][Tf₂N] cation and anion by a factor of two, while increasing the number of charge carriers in the gel.²⁹ The combination of MD simulations and NMR studies was used by Feng and co-workers³⁴ to study the differences between carbon and silica ionogels containing [bmim][Tf₂N]. The study showed that when confined in a silica matrix [bmim][Tf₂N] had smaller diffusion coefficients in the silica surface layer compared to the carbon surface layer. This observation was attributed to stronger interaction potentials between the silica and IL.³⁴ Recently, relaxation measurements of 10 wt. % plant based cellulose ionogels were shown to have two layers of diffusive motion where [bmim][Cl] relaxation was decreased resulting from changes in rotational behavior when confined.⁴⁸ It was recently shown in mesoscopically confined [emim][Tf₂N] in polyether sulfone membranes that the influence of an interface not only effects short range IL structuring but also influences long-range (>100 nm) dynamics within the sample.⁴⁹ The long-range effects are important to this work since bacterial cellulose (BC) is a porous material having pore sizes ranging from nano- to microscale dimensions, which suggests that any confinement effects occurring would be of a similar mesoscopic confinement.^{50,51}

Recently, bacterial cellulose ionogels (BCIGs) were prepared for the first time and their properties studied.⁵² Surprisingly, bacterial cellulose (BC) was found to exert a confinement effect on the [bmpy][Tf₂N] and [emim][Tf₂N] as determined by the change in the thermal crystallization and melting temperatures, T_{cr} and T_m , respectively, as well as peak broadening of the NMR spectrum.²³ As a supporting measurement, fluorescent monitoring of 1,3-bis(1-pyrenyl)propane showed change in internal parameters that are associated with viscosity and dynamics inside the BCIG. Despite the usefulness of these methods in discerning the presence of confinement inside the ionogel, these methods lend no detailed understanding to the nature of the confinement effects. The study of these gels is very intriguing for many reasons: first, the most widely reported ionogels are silica based;^{16,32} second, there have been no dynamical studies on ionogels containing >98 wt. % IL;³³ third, this is the first instance of dynamical studies of mesoscopic confinement inside a biopolymer; and finally, these materials have the potential to serve as greener alternatives in quasi-solid phase applications since they are made from BC, a high strength,⁵³ chemically stable,⁵⁴ biocompatible,⁵⁵ and highly tunable material.^{54,56} For these reasons, it is imperative to study gel mesoscopic confinement effects and the implications on future applications such as solid electrolytes, sensors, gas separation membranes, and catalysis platforms. Liquid-phase NMR and MD simulations were used to study the T_1 relaxation times, structure, and diffusivity of ILs confined within cellulose gels.

II. EXPERIMENTAL

A. Materials

Freeze-dried *Gluconacetobacter xylinus* bacteria (strain ATCC 700178) were obtained from ATCC (Manassas, VA).

Yeast extract, peptone, and dextrose were purchased from Fisher Scientific (Waltham, MA). Sodium phosphate dibasic and citric acid were acquired from Sigma-Aldrich (St. Louis, MO). Mannitol was purchased from Calbiochem Merck (Billerica, MA) and 200 proof ethanol was obtained from Decon Laboratories (King of Prussia, PA). Ionic liquids (ILs) were synthesized in-house using previously reported methods.⁵⁷

B. Bacterial cellulose hydrogel and alcogel growth and preparation

Bacterial cellulose (BC) was grown using bacterial strain *Gluconacetobacter xylinus* ATCC 700178, in a modified Hestrin Schramm medium similar to previous work.⁵² This medium consisted of a 2% w/v mannitol, 0.5% w/v yeast extract, 0.5% w/v peptone, 0.25% w/v sodium phosphate dibasic, and 0.15% w/v citric acid. The medium was adjusted to pH = 6 by adding 6M HCl and then autoclaved at 125 °C followed by the addition of 1% v/v of ethanol to the media using a sterile 0.2 μm syringe filter. Starter cultures were prepared by adding 5% v/v of mother media to fresh media. Cultures were allowed to grow for four days, at which point the first pellicle was removed from the surface with sterile forceps and discarded. Second pellicles were grown to desired thicknesses, harvested with sterile forceps, and then washed in a 1% NaOH bath at 95 °C for 1 h. After the NaOH washing, pellicles were transferred to a deionized water bath (~500 mL) where they were rinsed, changing the water every 2 h until a neutral pH was obtained. The cleaned hydrogels were then soaked in a 200 proof ethanol bath (100 mL) and changed every 2 h. The obtained alcogels were then stored until further use.

C. Bacterial cellulose ionogel (BCIG) preparation

BCIG samples for NMR measurements were prepared by soaking 2 cm × 2 cm × ~4 mm thick BC alcogels overnight in mixtures of ILs, [bmpp][Tf₂N] or [emim][Tf₂N] (~200% w/w IL/BC alcogel) and ~1 mL ethanol. The vials were capped, vortexed to mix ethanol and IL, and allowed to equilibrate overnight. The following day, the vials were uncapped and left open for ethanol evaporation for 2–7 days depending on atmospheric conditions. Following ethanol evaporation, the samples were weighed and measured for their dimensions and allowed to sit under vacuum for 12 h and were weighed and measured again. After BCIGs had finished drying, each gel was then cut, using a razor blade, into thin slices the length of the gel. The resulting slices were packed in an NMR tube, removing air bubbles when necessary.

D. NMR

No deuterium (No-D) NMR measurements^{58,59} were made on an Oxford AS600 NMR magnet with a Bruker AVIII HD 600 MHz console using a 5 mm CPTCI cryo-probe. ¹H and ¹⁹F-NMR were performed without spinning for both ionogels and their corresponding neat IL. The spectra were then calibrated using an external deuterated 4,4-dimethyl-4-silapentane-1-sulfonic (DSS) acid in D₂O standard for ¹H and hexafluorobenzene (C₆F₆) standard for ¹⁹F.

1. T₁ relaxation

No-D T₁ relaxation measurements were collected at temperatures ranging from 249 to 298 K using the Oxford AS600 instrument. The standard inversion-recovery pulse sequence was used with variable delays between 0.001 and 10 s randomly applied. Recycle delay was set to a value of 5*T₁. All spectra were phased by using the longest delay spectrum. The intensity of each peak was plotted over the series of delays and the resulting curve was fitted [Eq. (1)] using the Bruker TopSpin 3.5pl6 software. These measurements were performed for three samples of each gel and neat IL to obtain standard deviations as a function of temperature,

$$I[t] = I_0 + P * \exp\left(-\frac{t}{T_1}\right). \quad (1)$$

The fitted T₁ relaxation times were used to calculate the reorientation correlation times (τ_c) in the Bloembergen, Purcell, and Pound (BPP) theory [Eq. (2)], where ω₀/2π is the Larmor frequency, μ₀ is the vacuum permeability, γ is the gyromagnetic ratio of the proton or fluorine nuclei, ħ is the reduced Planck's constant, b is the distance between nuclei and the summed over those nuclei that are dipolar coupled to the nuclei in question, and I is the nuclear spin number,^{60,61}

$$\frac{1}{T_1} = \frac{6}{5} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma^4 \hbar^2}{b^6} I(I+1) \left[\frac{\tau_c}{1 + \omega_0^2 \tau_c^2} + \frac{4\tau_c}{1 + 4\omega_0^2 \tau_c^2} \right]. \quad (2)$$

Just as in previous studies, 0.616 = ω₀τ_c was used at the T₁ minima to evaluate the rotational correlation time (τ_c = 2.45 × 10⁻¹⁰ s) at the minimum.⁶⁰ With this evaluation of τ_c, b is calculated and is considered constant over the range of temperatures studied, with this value used to calculate all other τ_c values.

In this work, ¹H–¹H and ¹⁹F–¹⁹F dipole–dipole interactions were the only interactions of concern, ignoring the ¹H–¹⁹F interactions because these particular interactions are primarily dominated by translational and not rotational dynamics. In addition, ¹H–¹⁴N interactions were also neglected because of the small gyromagnetic ratio of the ¹⁴N nuclei (1.9338). This is shown by comparing Eqs. (3) and (4), which show that the contribution of the ¹H–¹⁴N interaction is negligible compared to the ¹H–¹H interaction,

$$^1\text{H}-^{14}\text{N} : \frac{\gamma_H^2 \gamma_N^2}{b^6} = \frac{26.75^2 \cdot 1.93^2}{b^6} = \frac{2678}{b^6}, \quad (3)$$

$$^1\text{H}-^1\text{H} : \frac{\gamma_H^4}{b^6} = \frac{26.75^4}{b^6} = \frac{512\,198}{b^6}. \quad (4)$$

2. Diffusion coefficients (D and D_{app})

The diffusion coefficients were measured with DOSY-NMR, using a gradient spin echo pulse sequence available through the TopSpin Bruker software similar to that previously reported by Stejskal and Tanner.⁶² Gradient strength was varied in 16 linear steps from 0.45 to 45.74 G cm⁻¹ using the SMSQ10.100 smoothing method, rectangular in shape with smoothed edges to give a gradient integral of 90% of the square pulse. The stimulated echo height decays were fit using

the TopSpin software according to Eq. (5), where $I(g)$ and I_0 are the echo heights at gradient strengths g and 0, respectively, D is the diffusion coefficient, γ is the gyromagnetic ratio of ^1H or ^{19}F nuclei, δ is the gradient-pulse length, and Δ is the diffusion time in the experiment. Self-diffusion within the ionogel (D_{app}) is an average of the dynamical processes occurring at the surface of the supporting matrix and the motions in the bulk-like regime. The δ (P30 in the Bruker software) value was kept constant at 2 ms for all experiments. Values for Δ in all systems were optimized over the range of 100–300 ms,

$$I(g) = I_0 \exp \left[-D(2\pi\gamma g\delta)^2 \left(\Delta - \frac{\delta}{3} \right) 1e4 \right]. \quad (5)$$

E. Molecular dynamics simulations

The large structural features (10–100 nm cellulose ribbon cross sections, nano- to micro-volume cavities)^{50,51} that are present in the investigated ionogel systems are beyond the reach of the molecular dynamics simulations that employ well-established all atom force fields. To benefit from the atomistic resolution and the compatibility of the Canongia Lopes and Padua (CL&P)^{63,64} and Optimized Potentials for Liquid Simulations (OPLS)⁶⁵ force fields, modeling smaller systems was chosen, where a bundle of cellulose is solvated in the ILs. Although the overall structure of the ionogel cannot be directly understood through this model, the cellulose effects on the IL dynamics can be observed by comparing the measured results to the pure IL simulations. However, IL cation and anion charge transfer effects observed in *ab initio* and density functional theory (DFT) calculations result in a decrease of the total ion charges to a value of $\pm 0.8e$.^{11,66–70} Accordingly, simulations that employ charges of $\pm 1e$ predict significantly stronger binding between the IL ions, and therefore the dynamics become systematically slower, underestimating diffusion constants while overestimating viscosities and lifetimes. To address these issues, partial atomic charges in the IL were scaled by a factor of 0.8^{70–72} which results in significantly better dynamic results.

Classical molecular dynamics (MD) simulations were performed with the LAMMPS⁷³ software. The periodic cubic boxes contained ca. 720 IL ion pairs and a bundle of β -cellulose composed of ten chains with eight β -D-glucose per chain or 260 ion pairs for the neat ILs. The boxes were created using the PACKMOL⁷⁴ program to have a density of 1 g cm^{-3} . After an initial energy minimization, a 1 ns long simulation was performed in the NpT ensemble ($T = 298 \text{ K}$, $\tau_T = 100 \text{ fs}$, $p = 1 \text{ bar}$, and $\tau_p = 1000 \text{ fs}$), where the volume of the box was averaged over the last 0.5 ns. This average volume was taken for subsequent NVT simulations. A 1 ns equilibration was followed by 50 ns of production run, where the coordinates were saved every 1 ps. A 1 fs time step was used throughout the entire simulation. Since the IL at 298 K was very viscous according to the current model and exhibited very slow dynamics, collecting the required data for lifetimes and rotational dynamics would have required an order of magnitude longer simulation time. To overcome this obstacle, we performed another set of simulations at 400 K, assuming that the trends in the dynamics would not be changed, while the faster

particle movement enables sufficient sampling for the calculation of dynamic properties. The trajectories were analyzed by the Travis⁷⁵ software.

III. RESULTS

A. NMR

The resulting ^1H and ^{19}F NMR spectra for [bmpy][Tf₂N] and [emim][Tf₂N] (Fig. 1) are presented in Figs. 2 and 3, where the spectra are the same in chemical shift of the neat and BCIG liquids, apart from the presence of the larger peak broadening in corresponding BCIG samples. Peak broadening accounts for multiple states of relaxation in the IL/BCIG system that is not present in the neat IL. Broadening found in both protons and fluorine atoms suggests that the confinement effects of the cellulose on the IL are acting on both the cation and the anion. The NMR spectral assignments were as follows:

[emim][Tf₂N] ^1H NMR (no-D) δ ppm relative to external standard DSS in D₂O: 1.37 (t, 3H, N—CH₂—CH₃), 3.77 (s, 2.99H, N⁺—CH₃), 4.10 (q, 1.98H, N—CH₂—CH₃), 7.27 (s, 0.99H, N—CH=CH—N⁺), 7.35 (s, 0.96H, N—CH=CH—N⁺), and 8.41 (s, 0.95H, N—CH=N⁺), [emim][Tf₂N] ^{19}F NMR (no-D) δ ppm relative to external standard C₆F₆: −78.64 (s, C—F), [bmpy][Tf₂N] ^1H NMR (no-D) δ ppm relative to external standard DSS in D₂O: 0.917 (t, 3H, N—(CH₂)₃—CH₃), 1.33 (m, 1.95H, N—(CH₂)₂—CH₂—CH₃), 1.70 (m, 1.94H, N—CH₂—CH₂—CH₂—CH₃), 2.15 (m, 3.99H, N—CH₂—CH₂—CH₂—CH₂—N), 2.96 (s, 2.96H, N—CH₃), 3.26 (m, 1.95H, N—CH₂—(CH₂)₂—CH₃), and 3.44 (m, 4.02H, N—CH₂—CH₂—CH₂—CH₂—N), [bmpy][Tf₂N] ^{19}F NMR (no-D) δ ppm relative to external standard C₆F₆: −78.28 (s, C—F).

1. T₁ relaxation

As seen in previous results⁵² as well as this study, the ^1H NMR of BCIGs displayed confinement effects through spectral broadening (Figs. 2 and 3). To examine this more

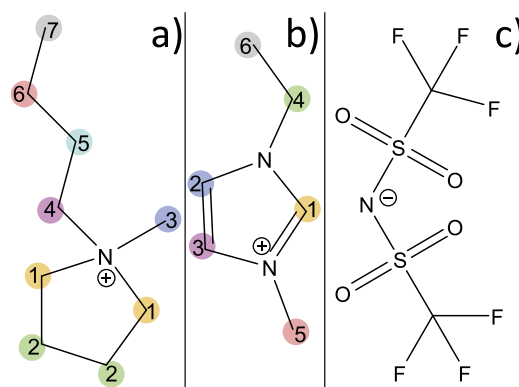


FIG. 1. Chemical structures of the cations (a) [bmpy]⁺ and (b) [emim]⁺ and (c) the anion [Tf₂N][−] used in this study. Cations are labeled in accordance with the measured ^1H NMR spectra starting downfield and are color-coded in the subsequent plots.

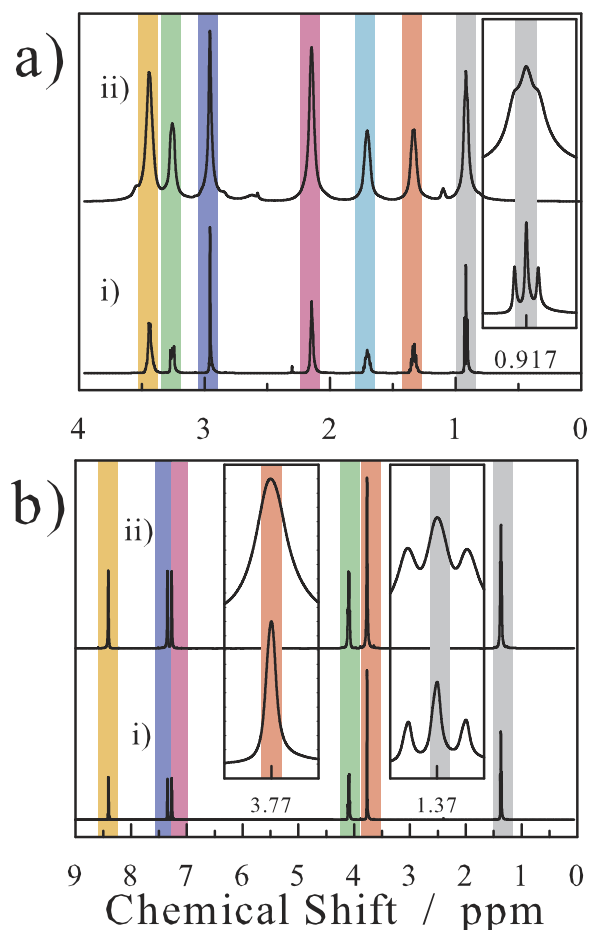


FIG. 2. No-D⁵⁸ ^1H NMR of (a) [bmppy][Tf₂N] and (b) [emim][Tf₂N] for (i) neat IL and (ii) ~99 wt.% BCIG samples. Insets are zoomed in spectra of individual peaks, highlighting the broadening that occurs in the spectra when the IL is confined in the BC.

closely, T_1 relaxation measurements for each hydrogen in the [bmppy]⁺ cation were collected between 249 and 298 K [Figs. 4(a) and 4(b) and Table S1 of the [supplementary material](#)] and between 250 and 298 K for [emim]⁺ [Figs. 4(c) and 4(d) and Table S2 of the [supplementary material](#)]. ^1H T_1 relaxation data collected for both ILs had similar trends to literature data for the high temperature side of the T_1 minima.^{76–78} These data were also consistent with the neat [bmppy]⁺ and [emim]⁺ IL viscosities of 76 mPa s¹³ and 30 mPa s,⁷⁹ respectively. T_1 relaxation is sensitive to local environments of nuclei, which can be indicative of site-specific site interactions. This was evident from the T_1 relaxation times of the BCIG confined ILs, where mesoscopic confinement of [bmppy]⁺ and [emim]⁺ showed decreased T_1 relaxation times for all IL protons. This decrease was indicative of changes in dipole-dipole interactions. In both ILs, the methyl substituents tended to be the most affected as evidenced by the T_1 relaxation time ratios of the IL and the BCIG at 298 K (see Table S3 of the [supplementary material](#)). The [bmppy]⁺ cation showed a ratio of ~1.7 which is most similar to the ratio at position 4 (~1.5, see Fig. 1). For the [emim]⁺ cation, the methyl group had a measured ratio of ~3.4 compared to the methyl group at position 6 (~1.7). Interestingly, when examining the ratios over the experimental temperature range, the methyl groups

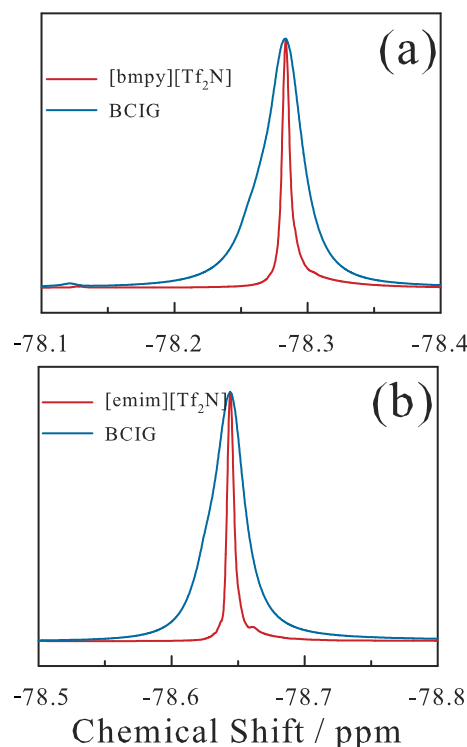


FIG. 3. No-D ^{19}F NMR of neat liquids and BCIGs for (a) [bmppy][Tf₂N] (chemical shift -78.28 ppm) and (b) [emim][Tf₂N] (chemical shift -78.64 ppm).

in both samples (position 3 [bmppy]⁺ and position 5 [emim]⁺) along with their positions 4 in [bmppy]⁺ and 6 in [emim]⁺ were the only ratios to change by more than 0.2. This observation suggested that the confinement in BCIGs started at the methyl position where the cellulose oxygens more readily formed hydrogen bonds, which is supported in MD simulations. The T_1 relaxation ratios also allow for a direct comparison of the cations studied. The ratios of each methyl group, position 3 in [bmppy]⁺ and 5 in [emim]⁺, at 298 K show that the [emim]⁺ cation has a higher affinity (>2.5 times) for interaction than the [bmppy]⁺ cation. The [emim]⁺ charge delocalization makes the entire molecule a charged species with little alkyl character as has been shown for shorter side chained ($n \leq 4$) imidazolium ILs, resulting in preferential interactions at sites 1 and 5.^{80,81} The [bmppy]⁺ cation does not benefit from charge delocalization and therefore is limited by its alkyl character. In each cation, the second highest ratios were observed at position 4 for [bmppy]⁺ and position 6 for [emim]⁺. The [bmppy]⁺ position 4 proton is opposite from the methyl group around the localized positive charge (see Fig. 1), so as the [bmppy]⁺ cation interacts with the cellulose chain, position 4 increases its interaction with the anions that were not primary interaction positions in the bulk systems. For [emim]⁺, the position 6 protons are pushed away from cellulose like the position 4 protons in [bmppy]⁺ and also interact with the anion more than expected, but this position also interacts with the cellulose surface as well. The difference between the cations is further illustrated when comparing methyl group ratios over the temperature ranges measured (Table S3 of the [supplementary material](#)). The [bmppy]⁺ position 3 methyl group relaxation shows a subtle linear decrease with temperature,

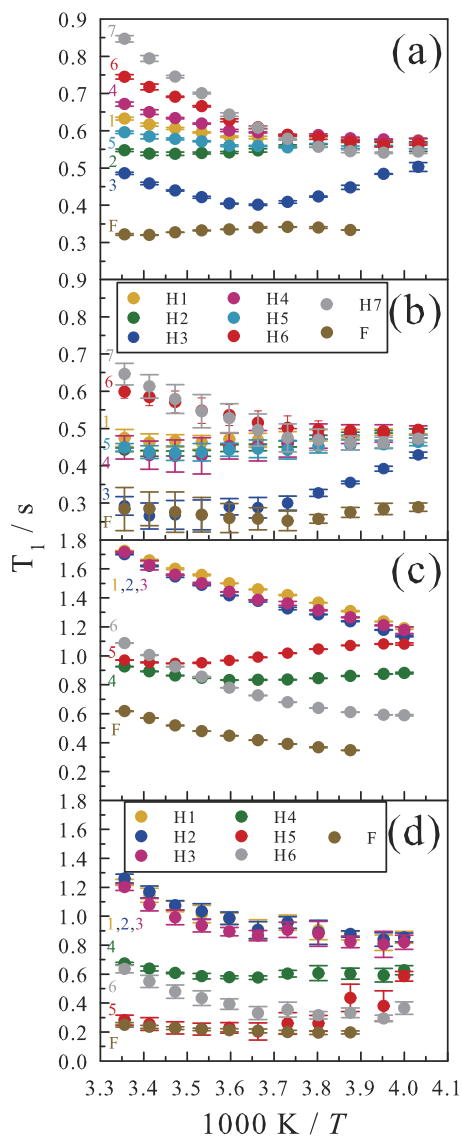


FIG. 4. Temperature dependence of the ^1H and ^{19}F T_1 relaxation times measured for (a) neat $[\text{bmpp}][\text{Tf}_2\text{N}]$, (b) $[\text{bmpp}][\text{Tf}_2\text{N}]$ BCIGs, (c) neat $[\text{emim}][\text{Tf}_2\text{N}]$, and (d) $[\text{emim}][\text{Tf}_2\text{N}]$ BCIGs.

while the $[\text{emim}]^+$ cation displays an increase in the ratio up to 4.9 ± 1.4 at 273 K and then proceeds to fall to 1.8 ± 0.1 at 250 K, which appears to follow a second order polynomial function. However, given the error bars associated with these ratios, at best it can only be stated that the ratios increase from around 1.8 ± 0.1 at 250 K to 4.9 ± 1.4 at 273 K and then decrease to 3.4 ± 0.4 at 298 K (which may not actually be a 2nd order polynomial trend). More data points are required to increase certainty with respect to the existence of a definitive trend. Nonetheless, the data do suggest that as the temperature is decreased, the interactions around the methyl group are reduced significantly. From the data, the primary interaction site of both ILs with cellulose is the cation methyl groups bonded at the positively charged nitrogen. In the $[\text{emim}]^+$ IL, this allows for anion positioning around the position 1 hydrogen, which agrees with previous reports which indicate that this location is a preferred anion interaction site.⁸⁰

When ^{19}F T_1 relaxation times were measured for the IL anion, differentiation was observed for both the neat $[\text{bmpp}]^+$

and $[\text{emim}]^+$ ILs, again coinciding with viscosities differences, 76 mPa s¹³ and 30 mPa s,⁷⁹ respectively. T_1 measurements of neat $[\text{emim}]^+$ yielded times that were larger than neat $[\text{bmpp}]^+$ due to the lower viscosity. However, this difference is changed when the ILs are incorporated inside the BC. Within the BCIG, $[\text{Tf}_2\text{N}]^-$ anion dynamics in the $[\text{bmpp}]^+$ IL statistically diverged from the neat IL dynamics just below 288 K, while in the $[\text{emim}]^+$ IL this anion diverged from the neat IL measurements at 298 K. The differences between the ILs inside the gels and the corresponding bulk liquids follow trends in the ^{19}F measurements similar to that observed with the ^1H measurements. When comparing the two liquids ratios, there are larger changes to the $[\text{emim}][\text{Tf}_2\text{N}]$ anion (ratio ~ 2.5 at 298 K) compared to the $[\text{bmpp}][\text{Tf}_2\text{N}]$ anion (ratio ~ 1.1). This large difference is due to the weak hydrogen bond acceptor being more attracted to the stronger delocalized positive charge of the $[\text{emim}]^+$ than the relatively localized positive charge of the $[\text{bmpp}]^+$. This is explained more thoroughly by ion pair lifetimes from the MD simulations.

Rotational correlation times (τ_c) were calculated and plotted against reciprocal temperature in an Arrhenius plot (Fig. 5) for each position on the IL cations, with τ_c values on the order of 50–600 ps over the range of temperatures (Tables S4 and S5 of the [supplementary material](#)). The calculated τ_c values for the $[\text{bmpp}][\text{Tf}_2\text{N}]$ cation showed that the H6 and H7 τ_c values did not change appreciably while the H1, H2, and H3 ring protons from the $[\text{emim}][\text{Tf}_2\text{N}]$ cation remained mostly unchanged. The protons in the $[\text{bmpp}][\text{Tf}_2\text{N}]$ cation displayed similar correlation time trends to previous measurements in the neat liquid.⁴² One of the τ_c value trends that was consistent throughout the $[\text{bmpp}]^+$ times was the butyl chain segmental motion, which showed decreasing values closer to the ring (H7 > H6 > H5 > H2). The $[\text{emim}]^+$ cation displayed the same segmental motion for its τ_c times (H6 > H4). The methyl groups on both the neat $[\text{bmpp}]^+$ and $[\text{emim}]^+$ cations displayed similar behavior to previous reports, where the pyrrolidinium cation methyl group has a faster τ_c than the other non-ring substituents, while the imidazolium cation methyl group displays a slower τ_c than all other groups.^{42–44} The $[\text{bmpp}]^+$ BCIG cation τ_c values, when calculated, did not change in the butyl chain, retaining the segmental motion of each proton in the chain. Within the $[\text{bmpp}]^+$ BCIG cation, however, the methyl group and the ring protons all had a slowing of their τ_c values, which suggested that the rotation of the molecule is slowing down when incorporated in the cellulose. This observation was corroborated in simulations. Interestingly, the $[\text{emim}]^+$ cation did not show major changes to τ_c for any of the ring protons or the ethyl chain, but did show an increase in the methyl group τ_c . This observation shows that the $[\text{emim}]^+$ cation hydrogen bonds to cellulose, with the interactions being localized to the methyl group.

Fluorine τ_c values for in the $[\text{bmpp}][\text{Tf}_2\text{N}]$ anion did not change to a large degree when subjected to confinement in cellulose. Remarkably, in the case of the $[\text{emim}][\text{Tf}_2\text{N}]$ anion, the fluorine τ_c values were shown to slow by approximately double when subjected to cellulose confinement. Again, this discrepancy is most likely due to the $[\text{emim}]^+$ IL having a hydrogen bonding capacity that is stronger than that of the

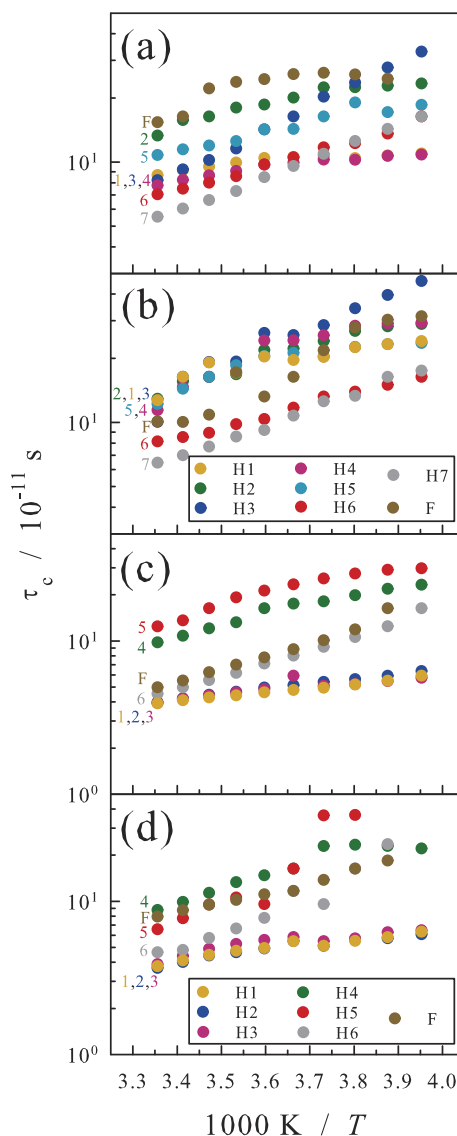


FIG. 5. Temperature dependence of the calculated rotational correlation times (τ_c) for each nucleus studied in (a) [bmpy][Tf₂N], (b) [bmpy][Tf₂N] BCIG, (c) [emim][Tf₂N], and (d) [emim][Tf₂N] BCIG samples.

[bmpy]⁺ IL, creating a more ordered environment around the cellulose.

2. D and D_{app}

Diffusion coefficients and apparent diffusion coefficients (D , D_{app}) were measured for both neat liquids and both BCIG samples to understand how the confinement of each liquid affects the translational dynamics within the samples and to better understand the rotational components of the relaxation measurements (Fig. 6 and Table S6 of the [supplementary material](#)). The measured D and D_{app} values for the [bmpy]⁺ cation were the same for every proton measured, showing no diffusional changes upon confinement. The D values for the neat [bmpy]⁺ cation matched well with the values measured previously,⁸² but unfortunately the authors are unaware of any studies measuring D_{app} in a confined system containing pyrrolidinium ILs without the presence of a ternary component like lithium. Despite this ternary additive, the trends for D_{app} seem

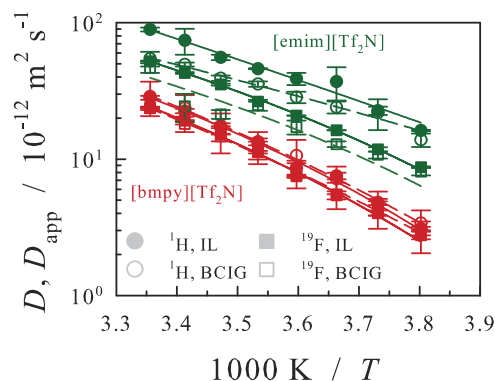


FIG. 6. Temperature dependence of the molecular diffusion coefficients (D) and apparent diffusion coefficients (D_{app}), for neat [bmpy][Tf₂N], neat [emim][Tf₂N] and BCIG samples fit with a standard Vogel-Fulcher-Tammann (VFT) approach over all nuclei and samples measured in this study.

to follow for the confined cation compared with ternary ionogels of [bmpy][Tf₂N]/Li⁺/[poly(diallyldimethylammonium)][Tf₂N].⁷⁷ D and D_{app} values from simulations clearly diverge from the experimental measurements, most likely due to the size of the simulated system being smaller than in the gel. Proton D and D_{app} measurements for the [emim]⁺ cation showed faster diffusion than the [bmpy]⁺ cation for both neat and confined ILs. The [emim]⁺ ¹H D measurements matched well with previous measurements of the neat [emim][Tf₂N] cation by Noda *et al.* and Tokuda *et al.*^{83,84} Confinement of [emim][Tf₂N] in BCIGs displayed similar effects to the liquid diffusion work performed by Han *et al.*, where [bmim][Tf₂N] was confined in mesoporous silica and D_{app} was decreased in the confined liquid.³² We are unaware of any existing measurements of [emim][Tf₂N] D_{app} values and thus cannot comment on direct comparison. The simulation and experimental data for [emim][Tf₂N] both follow similar trends unlike the [bmpy][Tf₂N] data, suggestive of long range motions present in the [emim]⁺ BCIG similar to those seen in polysulfone/[emim][Tf₂N] ionogels.⁴⁹ The discrepancy between the two cations is attributed to the charge localization in the [bmpy]⁺ cation, allowing apolar segregation in the liquid, while for the [emim]⁺ cation the molecule has delocalized charge that results in almost no alkyl character, thus preventing apolar segregation.

Diffusion measurements of the ¹⁹F nuclei in neat [bmpy][Tf₂N] were found to be similar to the ¹H D and D_{app} measurements of both the neat and confined liquid, but statistically lower than that of the neat liquid. The measurements at 293 K are in good agreement with those previously measured,⁸² but we are unaware of any ¹⁹F measurements for comparison of D_{app} in this pyrrolidinium IL that do not contain Li⁺. When comparing to previous ternary gel systems, similar D_{app} value trends were observed.⁷⁷ Also, [pmpy][Tf₂N] had very similar D values⁴⁶ and given that trends in pyridinium ILs for ¹⁹F nuclei diffusion are similar to ¹H diffusion measurements, compared to the equivalent imidazolium measurements being further apart, our measurements with pyrrolidinium compare well with the trends in the imidazolium and pyridinium liquids.⁸³ ¹⁹F diffusion measurements in neat [emim][Tf₂N] match well with previous studies conducted on

the neat liquid where fluorine diffusion was approximately half that of the ^1H D measurements.^{83–85}

Figure 7 shows the linear relationship between D and T/η predicted by the Stokes-Einstein theory of diffusion, Eq. (6), where k_B is the Boltzmann constant, T is the absolute temperature, η is the viscosity, and r_s is the hydrodynamic or Stokes radius of the molecule. In Fig. 7, the viscosity of the neat IL was the value used on the x-axis in both the neat liquid and gel plots, similar to previous measurements made on silica ionogels.³² These data follow trends that are similar to completely filled silica ionogels, with deviation from the neat IL.³² The Stokes-Einstein plot in Fig. 7 along with the diffusion plot in Fig. 6 clearly shows that the diffusion rate of the [emim][Tf₂N] sample slows down at the surface of the cellulose, which is a result of hydrogen bond formation with the cellulose hydroxyl groups and is perpetuated throughout the gel, while [bmpy][Tf₂N] does not have long range interactions of this variety,

$$D = \frac{k_B T}{6\pi\eta r_s}. \quad (6)$$

Upon examination of $\ln(T_1)$ vs $\ln(D, D_{\text{app}})$ plots in Fig. 8, it is clear that the slope of proton and fluorine nuclei in the [bmpy][Tf₂N] samples is not the same, which would be expected for a liquid sample that has multiple nuclei with varying contributions of rotation and translation components. Interestingly, the trend observed in the neat [bmpy][Tf₂N] nuclei was not seen in the BCIG sample, which would suggest a large change occurring in the rotational properties of the confined liquid. Correlation of the slopes from [bmpy]⁺ cation

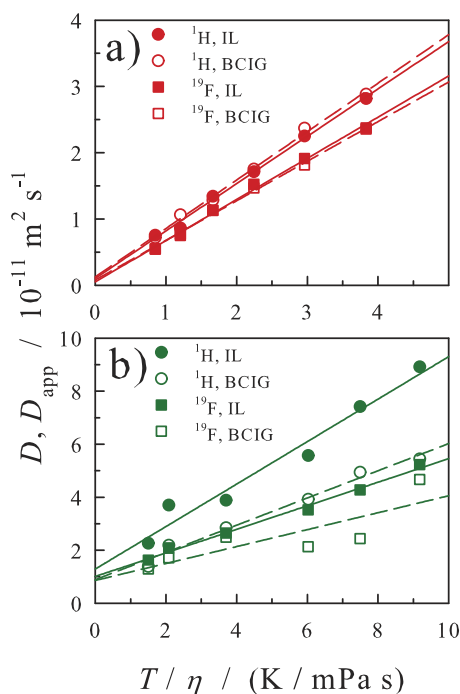


FIG. 7. Diffusion coefficients (D , neat liquid) and apparent diffusion coefficients (D_{app} , BCIGs) for (a) neat [bmpy][Tf₂N] and corresponding BCIGs and (b) neat [emim][Tf₂N] and corresponding BCIGs, plotted against T/η to demonstrate the conformity of the data to the Stokes-Einstein equation. Both ^1H and ^{19}F nuclei are included in the plots to show conformity of both cations and anion. The viscosity values, η , of the neat ILs were used for the viscosity values of the corresponding BCIGs.

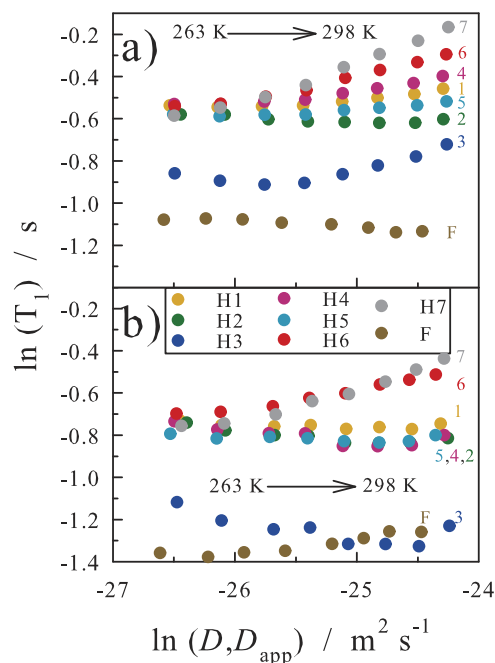


FIG. 8. Logarithmic plots of diffusion coefficients (D, D_{app} for ILs and BCIGs, respectively) and T_1 plots for (a) [bmpy][Tf₂N] and (b) [bmpy][Tf₂N] BCIG samples to elucidate the rotational and translational parts to each nuclei.

protons H1, H2, H4, and H5 showed that the T_1 relaxation measurements become largely translational, which could be indicative of a slowing down of the rotational dynamics. This is similar to results reported on fluorine dynamics in LiBF₄ in propylene carbonate where a larger rotational component was found.⁸⁶ The neat [emim][Tf₂N] plots in Fig. 9 show a similar result in discontinuity between slopes except in the case of the ring protons. The slopes for the [emim][Tf₂N] BCIG lost linearity, which suggested a change in the rotational dynamics for the [emim]⁺ sample as well. NMR relaxation measurements of imidazolium based ILs, in the presence and absence of confinement, appear to follow similar trends as our imidazolium samples with slower diffusion upon confinement resulting from a slower surface regime and changes in rotational dynamics that are concentration dependent.^{85,87,88} Upon examination of data obtained for [bmim][Cl] in a 10% cellulose ionogel, our data are supported and further show that at much higher concentrations these same relaxation and diffusion effects are seen in the IL.⁴⁸ These observations were corroborated by our MD simulations.

B. Molecular dynamics simulations

The liquid structure of the neat ILs was analyzed in detail to reveal the strongest interaction sites of the cation and the anion that may interact with the cellulose. We inspected the radial pair distribution functions between the nitrogen and oxygen atoms of the anion and each hydrogen atom of the cation. The sharpest and highest peaks at short distances, and therefore the strongest corresponding interactions, were provided by the oxygen atoms of the anion. These data are aligned with the findings from previous studies showing that the high degree of delocalization within the S-N-S unit disables the hydrogen bond acceptor ability of the imidic nitrogen atom (Fig. 10).

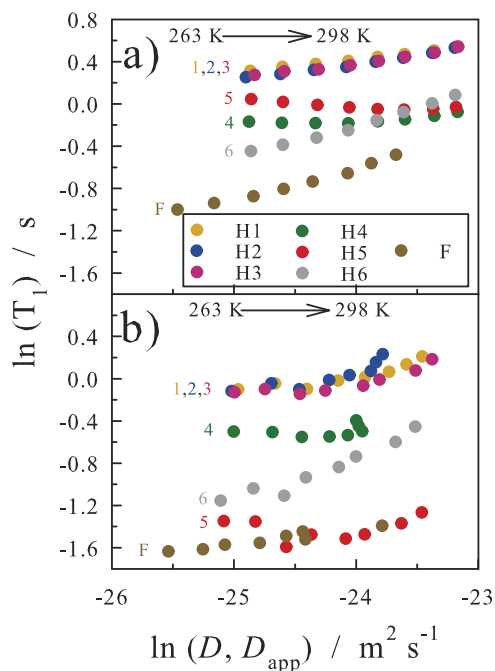


FIG. 9. Logarithmic plots of diffusion coefficients (D , D_{app} for ILs and BCIGs, respectively) and T_1 plots for (a) [emim][Tf_2N] and (b) [emim][Tf_2N] BCIG samples to elucidate the rotational and translational parts to each nuclei.

In the case of the [emim]⁺ cation, the dominance of the ring in hydrogen bond donor strength was clearly observed, especially at position 1 (Fig. 10), in clear accordance with previous data.^{80,89–92} The methyl group (position 5) also provides a notable peak, similar to that of position 3. This feature of imidazolium ILs has been recognized,⁸¹ and while the corresponding hydrogen bond is indeed detectable, the lack of large alkyl groups and the close proximity of the very strong position 1 interaction site that holds the anion at the same site

also contribute significantly to the stability of the bond. The ethylene unit (position 4) that is adjacent to the ring shows a notable but significantly lower peak, while the terminal methyl group at the side chain was the lowest (position 6), which has been shown before for ILs with this cation. For the [bmpy]⁺ cation, most of the peaks are somewhat lower, as one could expect from a structure where the interaction sites are all alkyl units. The strongest interaction sites are the methyl group and the ring methylene groups adjacent to the nitrogen atom, where the binding of the anion is not hindered by the alkyl groups with fluctuating conformations (Fig. 10), shown by the NMR measurements above. The hydrogen atoms become gradually less prone to interact with the anion toward the end of the butyl side chain, producing lower and lower peaks in the corresponding radial distribution functions (RDFs). This is in good qualitative agreement with the NMR experiments described above.

Apparently, both the cations and the anions interact with the cellulose through hydrogen bonding (Fig. 11). The importance of this dual interplay has been shown for the dissolution of cellulose in [emim][OAc] ILs.^{93–96} The corresponding RDFs show very low first peaks, which is due to the inaccessibility of cellulose building blocks in the middle of the bundle used in the normalization process to calculate the $g(r)$ values. For this reason, the height of these peaks is not an absolute measure of the interaction strength *per se*, but they still can be used for the comparison of the interactions between interaction sites within these two systems.⁹⁶ The [Tf_2N][−] anion is a weak hydrogen bond acceptor, but in the lack of competing acceptors in the IL, the anion forms hydrogen bonds with the cellulose. The anion-cellulose hydrogen bonds, according to the RDF peak heights, are somewhat stronger in case of the [bmpy]⁺ cation, where the cation is a weaker hydrogen bond donor, leaving the anion relatively free in the solution for further interactions. This trend is also visible in the anion-cellulose

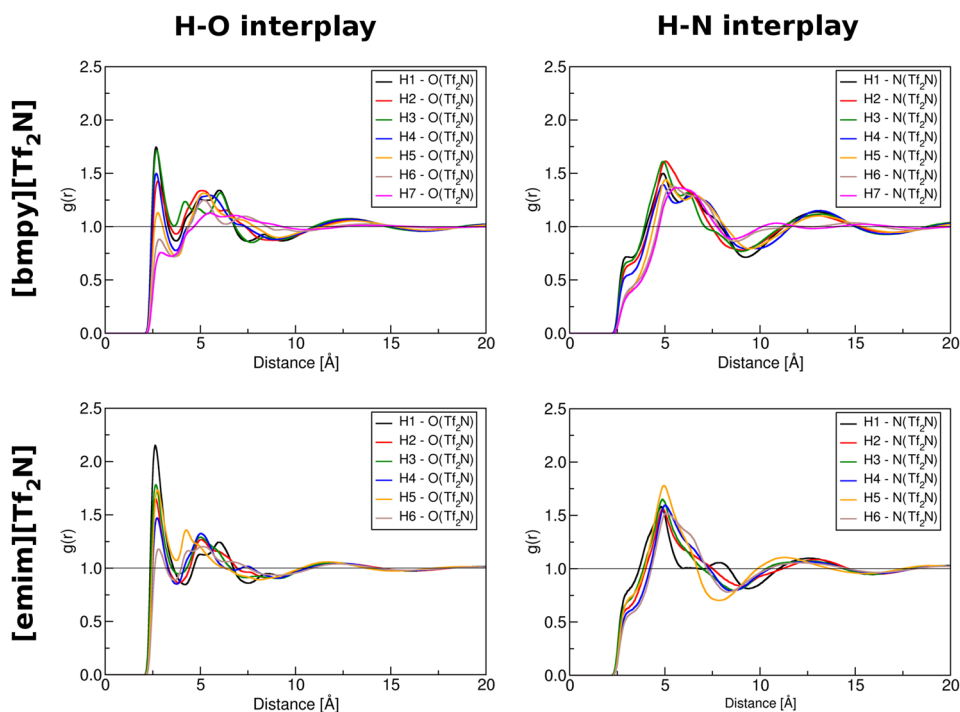


FIG. 10. Characteristic radial pair distribution functions for the conceivable hydrogen bonding partners in the two neat ionic liquids. The labeling of the hydrogen atoms is shown in Fig. 1.

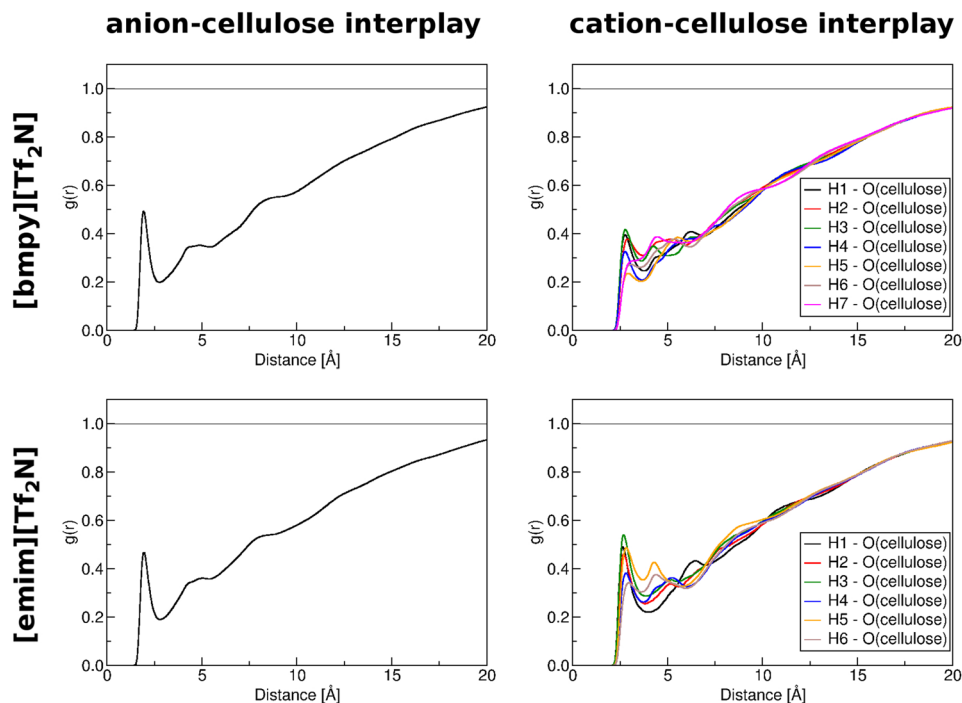


FIG. 11. Radial pair distribution functions' characteristic for the ionic liquid–cellulose interactions.

hydrogen bond lifetimes, which are—depending on the particular donor site of the carbohydrate—between 42 and 64 ps for the [emim][Tf₂N] and 105–140 ps for the [bmpy][Tf₂N]. On the other hand, the weak acceptor strength of the anion enables the cation to interact with the cellulose as well, producing close entries in the (low) first peaks in the corresponding RDFs.

It is interesting to observe that the order in the hydrogen bond donor strength changes for both the [bmpy]⁺ and [emim]⁺ cations compared to the observations regarding the anion-cation interplay in the neat liquids. For [emim][Tf₂N], the cation-anion interaction strength order in the neat liquid, H1 > H3 ≈ H5 > H2 > H4, changes to H3 > H1 ≈ H5 > H2 > H4 for the cellulose-cation interplay. The decrease in importance of the H1 position as an interaction site in the interplay with the cellulose can be explained through the competing strong anion-cation interactions. Since close approaches by the H4 and H6 to the cellulose are the rarest, these data allow concluding that the ethyl groups are generally pointing away from the surface of the cellulose. These results are fully consistent with the picture of a polar-polar interplay between the hydrophilic carbohydrate polymer and the ionic solvent. In case of the [bmpy]⁺ cation, the differences between the cation-anion and the cation-cellulose interplay concern mostly the butyl side chain. While the hydrogen atoms that are closest to the nitrogen exhibit the strongest correlation with the anions, decreasing gradually in strength toward the terminal methyl group, with the cellulose H5 having the lowest correlation, while H6 and H7 are significantly aligned more strongly to the cellulose. These peculiar results point out a delicate balance between the cellulose-cation, cation-anion, and anion-cellulose interactions, which determines the exact structure of the interfacial region.

The interactions described above should result in an ordered structure in the IL around the cellulose. Such layering

may have a considerable influence on the dynamics of the IL in the cellulose matrix, if its range is long enough to affect a large portion of the IL droplets absorbed within the BCIG. Defining the distance from the cellulose surface is not self-evident, due to the dynamic bending and other movements of the carbohydrate bundle. We chose three glucose units at the interface, which are farthest possible from each other, and their centers of ring provided the three points that were used to define the plane of the cellulose-IL interface. Naturally, due to the aforementioned movement of the cellulose block in the simulation box, the peaks are broad. The densities of the ions versus the distance of their centers of mass from the interface is shown in Fig. 12. In these plots, it is visible that the cations approach the interface closer in both cases, but the anions show, nevertheless, a higher first peak. This can be explained by the geometry of the ions, where the center of mass, considered in the distance calculations here, is closer to the interaction sites in the case of the cation than the stronger interacting anion. The second peaks at ca. 8–9 Å provide, for both ILs, a reverse order in

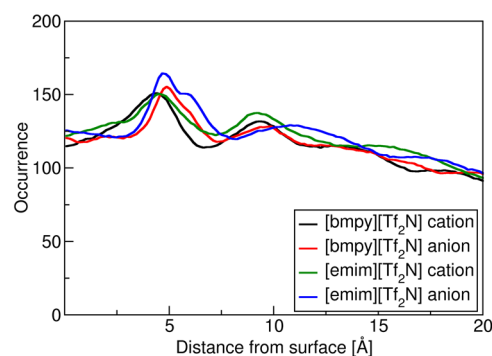


FIG. 12. Occurrences of the ions as a function of the distance from the cellulose–IL interface.

the peak heights, showing a higher probability of the cations than for the anions, which indicates that this peak arises from the interaction with the first layer. Accordingly, a picture of a bilayer structure emerges, where the surface of the cellulose is mostly covered by the anions, but a notable number of cations are also present, whereas the oppositely charged ions cover this first solvent shell. In both cases, the peaks dissipate after this second layer of ions, and after that at ca. 10–12 Å the density curves fall back to the base line, showing no further noteworthy effect on the IL. Thus, the liquid beyond that distance is practically unperturbed by the presence of the cellulose, and it should exhibit the properties of the bulk, which is mostly consistent with the very subtle changes observed in the experiments above.

The simulated diffusion constants of the neat ILs show the same trend as observed in the experiments, indicating a larger mobility for the [emim][Tf₂N] than for the [bmpy][Tf₂N] (Table I, cf. Fig. 6). In our simulations, the dynamics of the system slow down significantly in the presence of cellulose, resulting in diffusion constants being 30 (for [bmpy][Tf₂N]) and 50% (for [emim][Tf₂N]) of those in the neat systems (Table I). The lifetime of an ion pair—that is, the average time two oppositely charged ions spend as closest neighbors—is also significantly increased, indicating again slower dynamics in the presence of the carbohydrate. The vector reorientation autocorrelation functions relate to the rotational dynamics of the IL, showing how fast a given vector in the molecule at a given time is expected to fully lose its initial orientation. This can be observed in the decay of the underlying curves from a value of 1 to 0 in Fig. 13. It is visible from the curves that the rotational dynamics for the neat [emim][Tf₂N] are faster than for [bmpy][Tf₂N], in good agreement with the mobility data provided by the diffusion constants above. In all plots, the curves for the neat liquid (dotted lines) decay faster than the

TABLE I. Dynamic data on the simulated ILs. The ion pair dynamics were calculated by calculating the average time two oppositely charged ions spend as closest neighbors. The rotational correlation times were obtained as the double of the integrals of the curves in Fig. 13, which are reorientation autocorrelation functions for the defined vectors. In the case of the anion, only the N → S vectors were chosen, while for the [bmpy]⁺ cation the data are presented in the N → C4/N → C1/N → C3 order and for the [emim]⁺ cation the C1 → N(Me)/C1 → N(Et)/ring normal vector/C1 → H1 order.

	[bmpy][Tf ₂ N]		[emim][Tf ₂ N]	
	Neat	BCIG	Neat	BCIG
D_{anion} (10^{-12} m ² s ⁻¹)	210	73	297	190
D_{cation} (10^{-12} m ² s ⁻¹)	190	61	380	248
τ_{ionpair} (ps)	90	189	481	635
$\tau_{\text{rot,anion}}$ (ps)	167	505	71	144
$\tau_{\text{rot,cation}}$ (ps)	216/82/31	592/255/56	62/55/27/21	86/75/42/31

corresponding functions for the cellulose-containing systems (solid lines), showing the effect of the carbohydrate on the rotational dynamics. All the above measures of the dynamics indicate a more pronounced effect of the cellulose on dynamics for [bmpy][Tf₂N] than for [emim][Tf₂N]. For the numerical data with the time constants of the rotational movements—obtained by the double of the integral of the $c(t)$ curves in Fig. 13—see Table I. It is clearly visible from the data that the order in the rate of the different rotational movements is unchanged in the presence of the cellulose, showing that their rotation, apart from being slower, is generally very similar in the two environments.

When comparing the experiments and the simulations in this case, one must not forget that the systems that were investigated here by these two techniques are different. While the experiments included large domains of IL within the

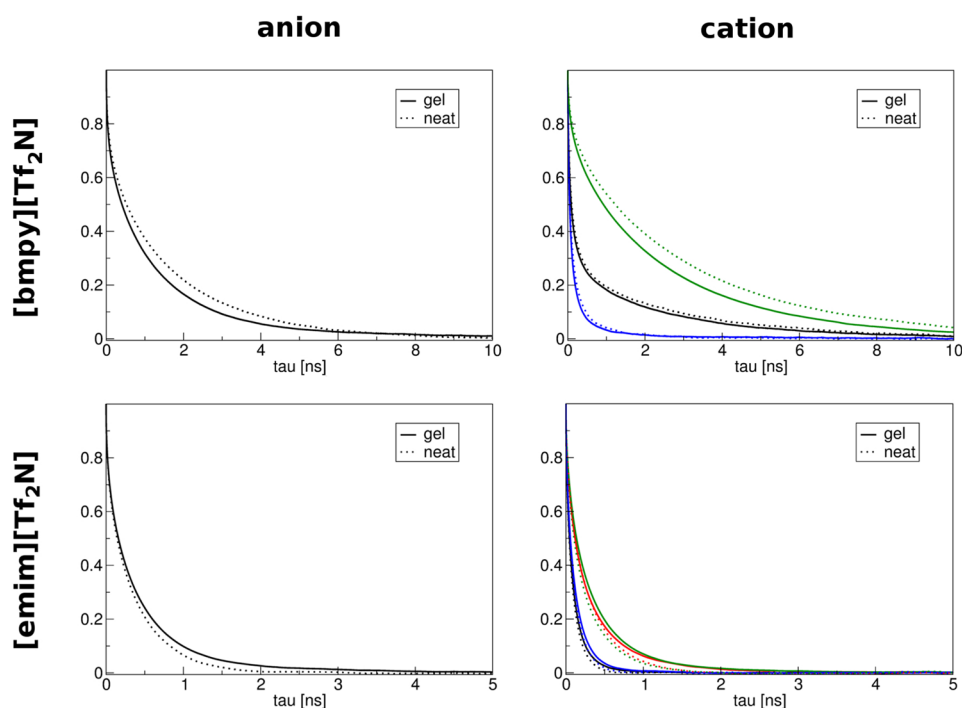


FIG. 13. Vector reorientation autocorrelation functions, where the rotational relaxation of the ions can be assessed. The chosen vectors for the anions were the N → S, for the [bmpy], the N → C1 (black), N → C3 (blue), and N → C4 (green), and for the [emim] the C1 → H1 (black), C1 → N (red and green), and the vector orthogonal to the ring plane (blue).

cavities of the cellulose, where the number of ions farther than 10–12 Å from the cellulose exceed the number of interfacial ions, the MD bulk/interface region ratio is significantly lower due to the smaller system size. In other words, the MD is more sensitive to the local changes in the structure and dynamics of the IL at the interface imposed by the presence of cellulose in the ionogel, than the experiments, which correspond more to the structure of the actual ionogel. Thus, the present results clearly indicate that while the cellulose has a strong local effect on the structure and dynamics of the liquid, the layered structure of the interfacial region does not reach far into the bulk, and therefore the large structural cavities in the BCIG should allow a large portion of the entrapped IL to behave predominantly as the bulk IL.

IV. CONCLUDING REMARKS

NMR measurements and MD simulations were used to study the structural, relaxation, and diffusion properties of the ionic liquids, 1-ethyl-3-methylimidazolium bis(trifluoromethyl-sulfonyl)imide, [emim][Tf₂N], and 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide, [bmpp][Tf₂N], in order to understand the mesoscopic confinement they undergo within bacterial cellulose ionogels (BCIGs). Despite very small amounts of cellulose being present (~1% by mass), and pore sizes ranging from nano- to microscale,^{50,51} experimental results show large decreases in the observed properties (e.g., T₁ relaxation times and diffusion coefficients) upon confinement of the liquid in the cellulose matrix. If these effects were only carried out as far as the liquid structural layering observed at the surface of the cellulose (~10–12 Å) in such large voids, the bulk-like dynamics of the liquid should surmount that of the interfacial layer. In the [bmpp][Tf₂N] system, regardless of the localized charge with sterically hindered interaction with the surface, strong cellulosic interaction was still observed through extended lifetimes of ion pairs and T₁ relaxation times decreasing by ~1.6 times the neat liquid. The [emim]⁺ cation had even larger effects on dynamic and relaxation properties, but showed half the hydrogen bonding lifetimes with the cellulose. The diffusion information obtained through DOSY-NMR and MD simulations showed that both ILs have areas in which their diffusion properties decrease, with [emim][Tf₂N] having the largest decrease observed in MD simulations and experiment. Generally, the effects of an interface on a molecular liquid dissipate after a short distance, giving a more bulk-like character to largely filled pores as was observed in the diffusional properties of [bmpp][Tf₂N]. In the case of [emim][Tf₂N], NMR experiments showed a greater degree of interfacial influence, which should not be observed with large amounts of liquid (~99 wt. %). This was suggestive of a long-range influence from the matrix on the liquid dynamics in [emim][Tf₂N], which has been previously reported in mesoscopic confinement of [emim][Tf₂N].⁴⁹ The MD simulations of [bmpp][Tf₂N] showed that the diffusional properties of the liquid at the interface were different than those of the bulk liquid which would suggest that in the large cavities of BC the alkyl chains forming apolar regions throughout the liquid dominates over the interface induced structuring of the IL that

only extends a short distance. Given the need for immobilizing ILs in solid supports that can then be used for various applications such as quasi-solid electrolytes, separation membranes, optically active materials, and drug delivery gels, this study lends a great deal of understanding to what occurs in ionogels to different types of ionic liquids that are mesoscopically confined. In the future, this study could be extended to other cation types, anion types, alkyl chain lengths, additives, and cellulose modifications.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for the tabulated data for T₁ relaxation measurements, T₁ relaxation ratios, rotational reorientation times, and diffusion measurements.

ACKNOWLEDGMENTS

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